

Week 7 Lab: MongoDB Aggregation Framework - Passenger Ride Analysis

Objective: Master MongoDB's aggregation framework using `$lookup` to analyze passenger ride data. Learn to design pipelines in MongoDB Compass and implement them in Node.js for the e-hailing system.

Lab Overview

Key Concepts

1. **Aggregation Pipeline:** `$match`, `$lookup`, `$unwind`, `$group`, `$project`
2. **Multi-Collection Joins:** Combine user and ride data using `$lookup`
3. **Compass Visualization:** Build and debug pipelines visually
4. **Node.js Integration:** Export pipelines from Compass to code

Deliverables

1. MongoDB aggregation pipelines for passenger analysis
 2. Node.js API endpoint with aggregation implementation
 3. Postman tests with sample outputs
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Lab Procedures

Part 1: Data Preparation

Create Sample Data

Insert multiple sample documents into `users` and `rides` collections based on the you designed in the Entity Relationship Diagram

Part 2: Pipeline Design in MongoDB Compass

Build Aggregation Pipeline

1. Open MongoDB Compass and connect to your database
2. Select the `users` collection → Aggregations tab
3. Create a pipeline to calculate the statistics of your passengers which return the following result:

```
[
  {
    "name": "Alice",
    "totalRides": 2,
    "totalFare": 37.8,
    "avgDistance": 10.35
  },
  {
    "name": "Bob",
    "totalRides": 1,
    "totalFare": 18.75,
    "avgDistance": 9.8
  }
]
```

Part 3: Node.js Implementation

Create API Endpoint

1. Implement `GET /analytics/passengers` endpoint based on the codes from the Mongo Compass Aggregation Pipeline you designed in Part 2.

Part 4: Postman Testing

Test with Postman

1. Create GET request to `http://localhost:3000/analytics/passengers`

Part 5: Analytic Dashboard

Export your all Postman API including the latest passengers analytic, and the produce a professional dashboard to illustrate the data (e.g. chart, figure, table).

Submission Requirements

1. GitHub Repository
2. Mongo Compass Aggregation Pipeline Design
3. Postman Test Result
4. Screenshots of the Dashboard